Exp 14: Exploring Pro-Code Extension of a Zoho Creator Application using Zoho Catalyst / Custom Functions

1. Aim

To design a Zoho Catalyst extension for the **Loan Application & Pre-Screening Management** app. The extension produces a structured, nonbinding pre-screening checklist from permitted application data and returns it to the loan officer workflow. Final eligibility and approval remain with authorized bank staff.

2. Procedure

Step 1: Define a narrow service boundary

Creator retains Loan_Application_Form, required-field validation, Loan_Application_Report, and acknowledgement. A serverless function receives a case reference, applicant age, Monthly_Income, Loan_Type, and Consent_For_Credit_Check status. It returns validation flags and a *manual review required* status. Do not transmit PAN_Number or full address to this function.

Step 2: Prepare Catalyst

Create a Catalyst project for the loan workflow, configure a function endpoint, and store authentication secrets in managed configuration. Restrict access to a protected Creator connection. Keep development and production endpoints separate.

Step 3: Implement deterministic checks

Check age 18 or above, income greater than zero, recognized loan type, and recorded consent. Return a case-scoped response with reasons for any failed check; do not call this a credit score or approval decision.

Step 4: Integrate and test

Invoke the endpoint after a successful form submission. Save only the checklist outcome and correlation ID. Test valid, invalid, duplicate, timeout, and unavailable-service cases; route failures to manual review without losing the application record.

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3. Output

3.1 Architecture division

Layer	Platform	Responsibility
Data and forms	Zoho Creator	Thirteen Basic Fields, validation, Loan_Application_Report
Workflow	Zoho Creator	Acknowledgement and loan officer review
Checklist API	Zoho Catalyst	Normalize request and return nonbinding flags
Audit	Bank-controlled logs	Correlation ID, policy version, and processing status

3.2 Example request and response

```
{
  "caseRef": "LA-EXAMPLE-001",
  "ageYears": 36,
  "monthlyIncome": 55000,
  "loanType": "Home Loan",
  "creditCheckConsent": true
}

{
  "caseRef": "LA-EXAMPLE-001",
  "status": "MANUAL_REVIEW_REQUIRED",
  "checks": { "adult": true, "incomePositive": true,
              "loanTypePresent": true, "consentRecorded": true },
  "policyVersion": "intake-v1"
}
```

The sample values illustrate the Main_ref scenario. The response confirms intake checks only; it contains no approval decision.

3.3 Catalyst function sketch (Node.js)

```
module.exports = async (req, res) => {
  const { caseRef, ageYears, monthlyIncome,
          loanType, creditCheckConsent } = req.body || {};
  if (!caseRef) return res.status(400).json({ error: "caseRef required" });
  const checks = {
    adult: Number(ageYears) >= 18,
    incomePositive: Number(monthlyIncome) > 0,
    loanTypePresent: typeof loanType === "string" &&
                     loanType.trim().length > 0,
    consentRecorded: creditCheckConsent === true
  };
  return res.status(200).json({
    caseRef,
    status: "MANUAL_REVIEW_REQUIRED",
    checks,
    policyVersion: "intake-v1"
  });
};
```

Deploy this behind authenticated access, request validation, rate controls, and structured error logging. Determine age in Creator from Date_Of_Birth before sending; the date itself need not leave the form service for this check.

3.4 Creator integration sketch

```
// Pseudocode: use an authorized Creator connection.
payload = Map();
payload.put("caseRef", input.ID.toString());
payload.put("ageYears", calculatedAge);
payload.put("monthlyIncome", input.Monthly_Income);
payload.put("loanType", input.Loan_Type);
payload.put("creditCheckConsent", input.Consent_For_Credit_Check);
response = invokeurl
[
  url: secureCatalystEndpoint
  type: POST
  parameters: payload.toString()
  connection: "loan_checklist_connection"
];
```

3.5 Test and failure matrix

Scenario	Expected response	Officer action
Valid intake	All checks true; manual review required	Verify documents and assess case
Under-age applicant	Adult check false	Do not proceed without policy review
No consent	Consent check false	Do not initiate bureau check
Zero or missing income	Income check false	Request correction
API unavailable	Record preserved; retry or error logged	Manual review queue
Duplicate callback	Same case reference recognized	Avoid duplicate downstream checks

4. Result / Observations

1. The extension separates custom processing from the form and report workflow while keeping the intake user experience simple.
2. Data minimization keeps PAN and address out of the Catalyst request.
3. Every response remains a pre-screening aid; approval, bureau processing, and policy exceptions require bank-controlled procedures.

Run the tests against a secured Catalyst deployment before claiming measured latency or successful production integration.

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